

StayEase Data Analytics Report

Data Analytics with Python

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Company	StayEase
Dataset	Student_16_Dataset.xlsx

Hotel Booking Analytics • Pandas • NumPy • Matplotlib • Seaborn

1. Dataset Overview

StayEase is the hotel-booking business domain assigned to Student_16. The analysis uses the provided Student_16_Dataset.xlsx file and follows the capstone structure: load and inspect the dataset, clean real-world data issues, perform exploratory analysis and NumPy operations, build visualizations, and convert the findings into business insights and recommendations.

The final cleaned analytical dataset contains 1,973 records and 20 columns. The available fields cover booking identification, dates, customer attributes, geography, category, quantity, pricing, discount, Sales, Profit, ServiceCharge, DeliveryDays, PaymentMode, and Rating.

Metric	Value
Records analyzed	1,973
Columns	20
Total Sales	₹70,949,243.67
Total Profit	₹14,672,092.73
Average Sales per booking	₹35,960.08
Median Sales per booking	₹23,374.49
Average Profit per booking	₹7,436.44

The capstone instructions define Sales as the final transaction amount and Profit as the profit earned on the transaction; they also identify Region/State/City, Category/SubCategory, Quantity, UnitPrice, Discount, PaymentMode and Rating as key analytical fields.

2. Data Cleaning Steps Taken

- Inspected the dataset structure using shape, columns, index, head, tail, info, describe, and missing-value checks.
- Handled missing CustomerName values as part of the cleaning workflow.
- Handled missing Rating values so that missing ratings did not prevent the remaining records from being analyzed.
- Identified and removed duplicate rows.
- Standardized inconsistent City text such as uppercase/lowercase variants (for example, Mumbai/MUMBAI) so city names were represented consistently.
- Converted OrderDate into a proper datetime representation for time-based analysis.
- Verified categorical values after cleaning; Category contained 5 valid categories with no remaining missing Category values.
- Kept the StayEase domain-specific ServiceCharge field as the applicable extra-fee column.

The capstone specifically requires missing values, duplicate rows, inconsistent city text, and date types to be identified and handled during Phase 3.

3. Key Statistics & EDA Findings

3.1 Sales and Profit

Total Sales were ₹70.95 million, while total Profit was ₹14.67 million. Average Sales was ₹35,960.08 compared with a median of ₹23,374.49, showing that higher-value bookings raise the mean above the median. Profit ranged from -₹3,330.73 to ₹51,042.55, with an average of ₹7,436.44.

3.2 Regional Performance

Rank	Region	Total Sales
1	South	₹19,017,458.06
2	East	₹18,910,717.64
3	North	₹17,061,550.75
4	West	₹15,959,517.22

3.3 Category Performance

Category	Bookings	Average Profit
Banquet/Event	379	₹16,449.37
Villa	405	₹10,239.92
Suite	401	₹6,462.32
Deluxe Room	389	₹2,793.11
Standard Room	399	₹1,535.59

3.4 Payment Behavior

Payment Mode	Bookings	Share
Wallet	411	20.83%
Net Banking	405	20.53%
UPI	394	19.97%
Debit Card	391	19.82%
Credit Card	372	18.85%

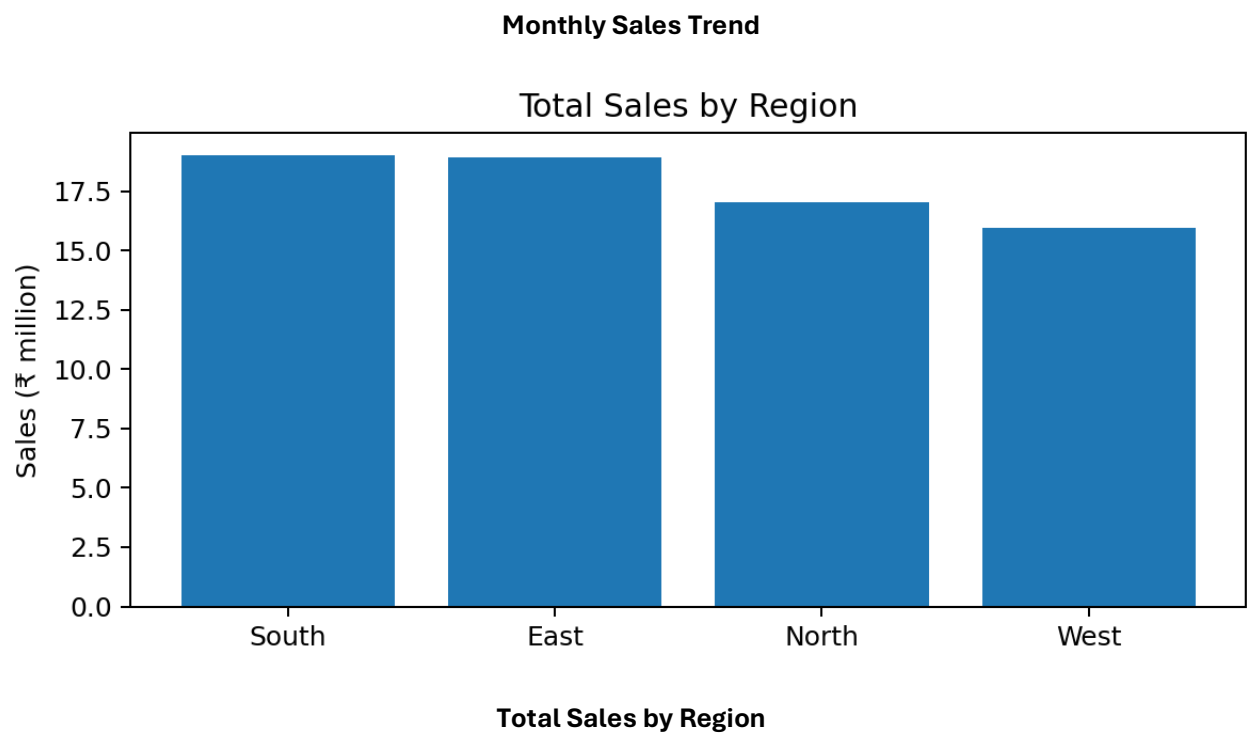
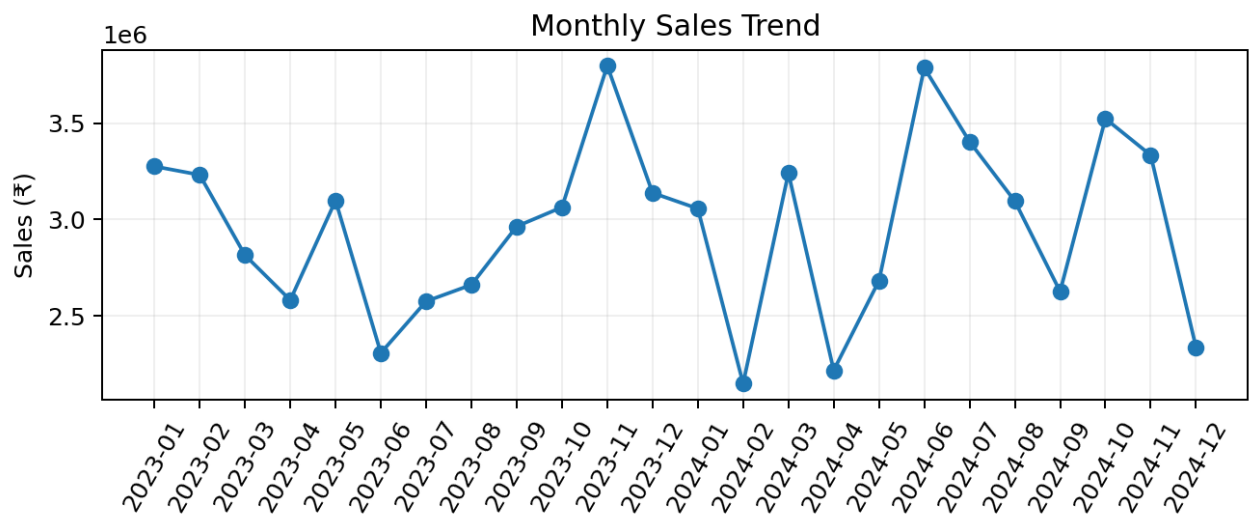
3.5 Correlation

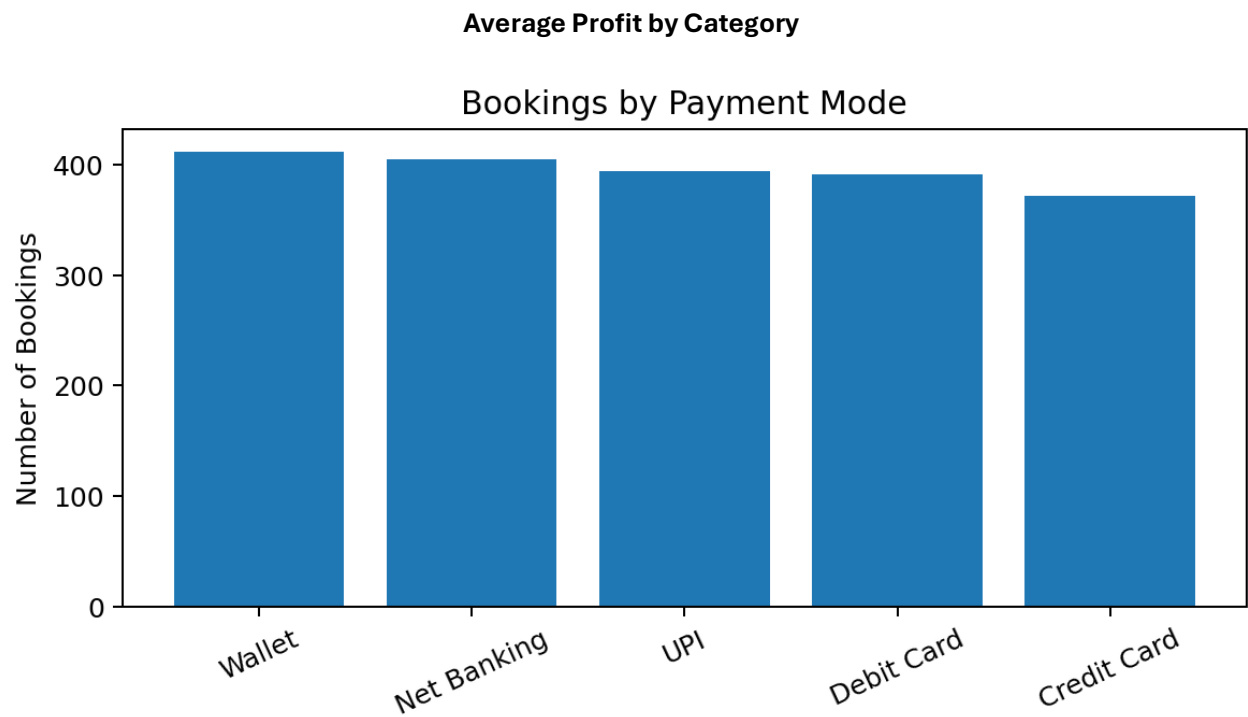
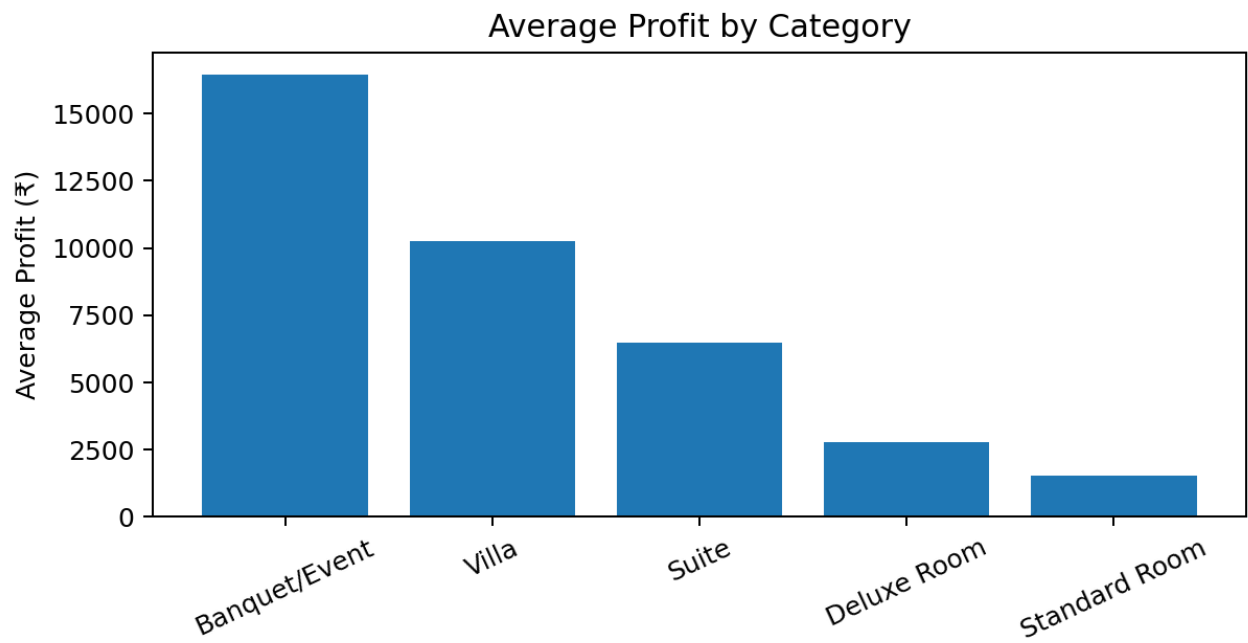
The strongest relationship among the analyzed numerical variables was between Sales and Profit, with a Pearson correlation of 0.8596. This indicates a strong positive linear association: higher Sales values generally occur alongside higher Profit values. Correlation is an association measure and does not by itself establish causation.

These analyses correspond to the capstone requirements for Sales statistics, correlation, PaymentMode frequency, unique Categories, GroupBy summaries, and Pivot Tables.

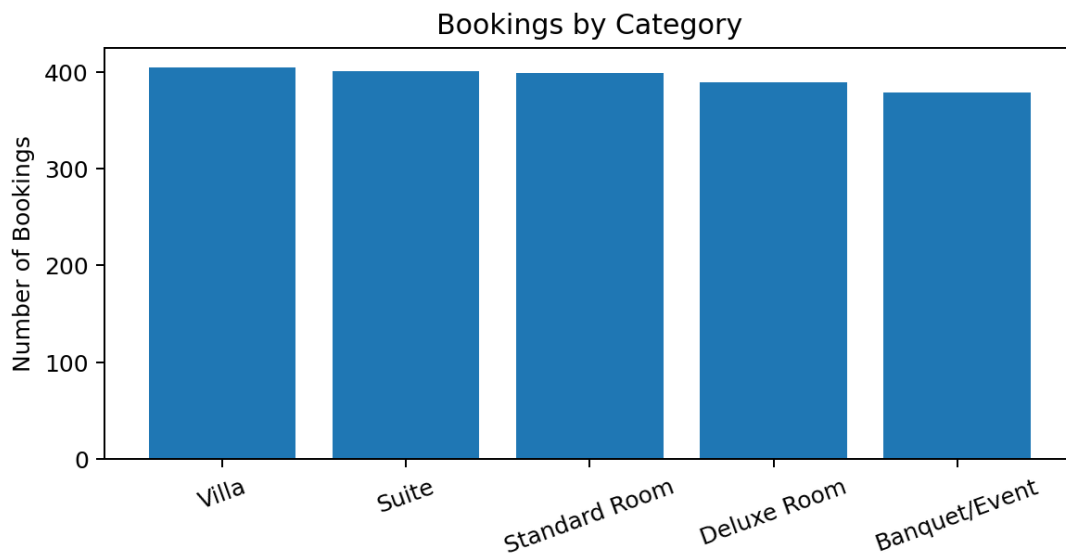
4. Dashboard & Visual Analysis

The report dashboard below uses six presentation-ready views built from the completed analysis. The full notebook also contains the additional boxplot, scatterplot, and correlation heatmap used during the visualization sprint. The capstone requires the final dashboard to combine 6–8 of the strongest charts and to give the dashboard and each chart clear titles.



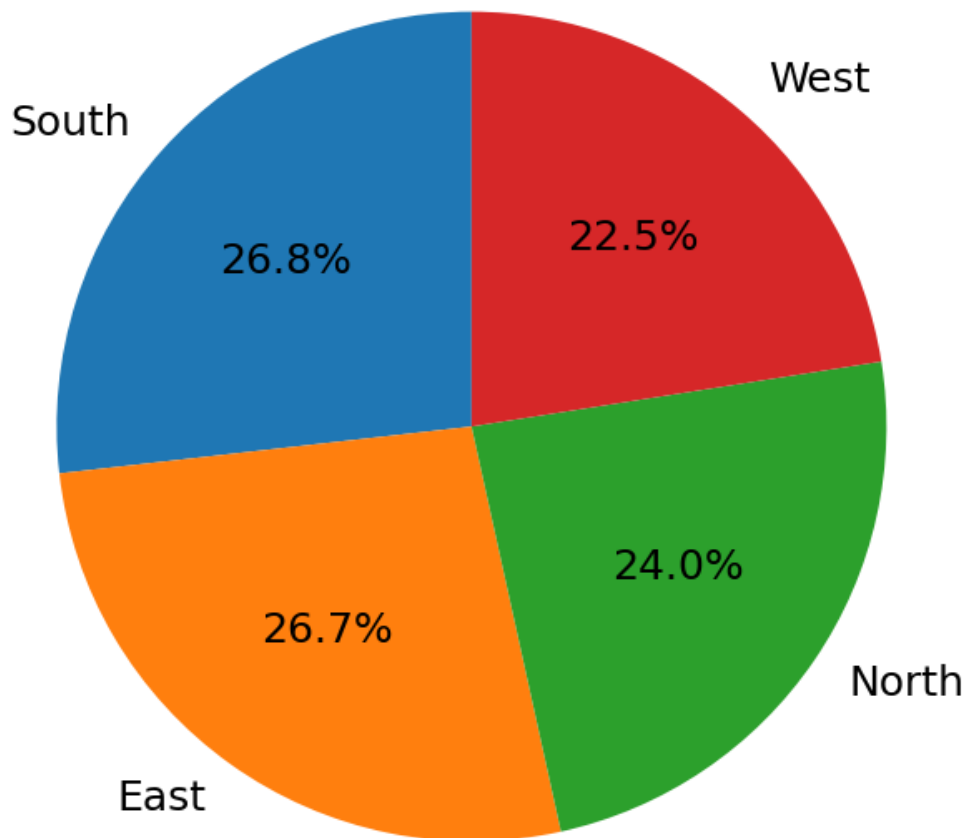


Bookings by Payment Mode



Bookings by Category

Sales Share by Region



5. Business Insights

- StayEase generated ₹70.95 million in total Sales across 1,973 bookings. Average Sales were ₹35,960.08 while the median was ₹23,374.49, indicating that higher-value bookings pull the average above the median.
- The South region generated the highest total Sales at ₹19.02 million, followed by East at ₹18.91 million, North at ₹17.06 million, and West at ₹15.96 million.
- Wallet was the most-used payment method with 411 bookings, representing 20.83% of all bookings. The payment mix was relatively balanced, with each of the five payment modes accounting for roughly 19–21% of bookings.
- Banquet/Event produced the highest average Profit at ₹16,449.37 per booking. Villa followed at ₹10,239.92, while Standard Room had the lowest average Profit at ₹1,535.59.
- Sales and Profit had a strong positive correlation of 0.8596, showing that higher Sales values are generally associated with higher Profit values in the dataset.

The capstone requires each business insight to be supported by a specific number from the analysis.

6. Recommendations

Recommendation 1 — Strengthen the South Region

The South region is the strongest regional contributor by total Sales at ₹19.02 million. StayEase should identify the properties, customer segments, pricing approaches, and sales practices contributing to this performance, then evaluate whether successful practices can be replicated in lower-performing regions.

Recommendation 2 — Focus on the Banquet/Event Category

Banquet/Event has the highest average Profit at ₹16,449.37 per booking. StayEase should investigate the pricing, demand, and cost structure behind this result and strengthen profitable event offerings where demand supports it.

7. Conclusion

The StayEase project demonstrates an end-to-end Python data analytics workflow. The dataset was inspected and cleaned, followed by statistical analysis with Pandas and numerical analysis with NumPy. Matplotlib and Seaborn were then used to turn the findings into presentation-ready visualizations and a consolidated dashboard.

The analysis identifies South as the highest-Sales region, Wallet as the most-used payment method, and Banquet/Event as the highest-average-profit category. The strong Sales–Profit correlation further shows a clear positive relationship between transaction value and profitability. These findings provide practical directions for regional strategy and category-level profitability.

8. Code & Submission Reference

The complete executable analysis is maintained in the Jupyter Notebook: StayEase_Analysis.ipynb. The notebook contains the data loading, cleaning, Pandas/NumPy analysis, Matplotlib and Seaborn visualizations, dashboard code, business insights, and recommendations.

The capstone instructions require the submitted write-up to contain Dataset Overview, Data Cleaning Steps Taken, Key Statistics & EDA Findings, Dashboard, Business Insights, and Recommendations, and to include the actual code or a link/attachment to the notebook.